

Assignment 1: Comparative Evaluation of Classification Models

Objective: This assignment evaluates your ability to apply and compare three classifiers **Logistic Regression**, **Decision Tree**, and **Support Vector Machine (SVM)** on datasets of different types of class boundaries (linear vs non-linear).

Datasets Provided

Two datasets are provided for this assignment:

- **Linear Dataset** : Download: `linear_dataset.csv`
- **Non-Linear Dataset** : Download: `nonlinear_dataset.csv`

Each dataset contains:

- Two numerical input features: `Feature_1`, `Feature_2`
- One binary output class label: `Target`

Task 1: Data Preparation

1. Load each dataset into Python using Pandas.
2. Display the first 10 rows along with summary statistics.
3. Generate a scatter plot of the two classes for each dataset.
4. Briefly comment on whether the dataset appears linearly separable.

Task 2: Model Training

Train the following classifiers **separately on each dataset**:

- Logistic Regression (default parameters)
- Decision Tree (`max_depth = 3`)
- Support Vector Machine:
 - `kernel='linear'`
 - `kernel='rbf'`

Task 3: Model Evaluation

For each trained model:

- Compute Accuracy
- Display the Confusion Matrix
- Report Precision, Recall, and F1-score
- Plot the corresponding Decision Boundary

Note: Use either `matplotlib` or `mlxtend.plotting` for decision boundary visualization.

Task 4: Interpretation and Analysis

Write a short structured explanation (8–12 sentences) addressing:

- Which models performed best on the **linear** dataset, and why?
- Which models performed better on the **non-linear** dataset, and why?
- Why does SVM with **rbf** kernel help in the non-linear case?
- Comment on differences between **macro** and **weighted** averages in evaluation metrics.
- Discuss which model's decision boundary is most interpretable and when this matters.

Expected Learning Outcomes

By completing this assignment, you will demonstrate the ability to:

- Recognize when a dataset is linearly vs non-linearly separable.
- Select appropriate classifiers based on separation characteristics.
- Interpret confusion matrices and classification reports meaningfully.
- Understand the role of kernel functions in SVM.
- Communicate machine learning results clearly and analytically.